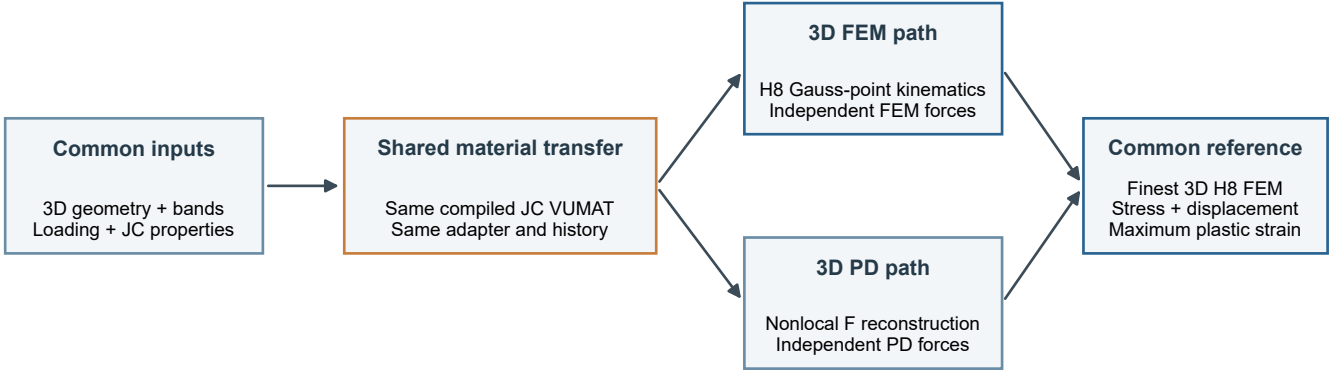Verification with matched three-dimensional models